

FORT WORTH ALLIANCE, TX, USA

WMO: 722594

Lat: 32.973N

Lon: 97.318W

Elev: 685

StdP: 14.34

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 53909

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	21.1	25.3	8.6	8.8	29.6	12.4	10.6	34.8	27.1	48.6	24.4	50.6	7.5	0	0.495

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	21.5	102.3	74.0	99.9	74.5	97.5	74.3	78.3	92.4	77.5	91.6	76.8	90.5	9.0	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.8	133.5	83.8	73.5	127.7	82.3	72.9	125.4	82.2	42.2	93.0	41.3	91.7	40.5	90.3	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.4	20.5	18.6	DB	14.5	105.6	3.8	3.4	11.7	108.1	9.5	110.1	7.3	112.0	4.5	114.4	(4)
			WB	13.1	80.1	3.6	1.4	10.5	81.1	8.4	81.9	6.4	82.7	3.8	83.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	66.2	45.8	49.6	57.4	65.0	73.4	81.8	85.6	85.7	78.2	67.2	56.2	46.9	(6)
		DBStd	16.38	10.11	10.56	9.78	7.78	6.73	4.69	4.32	5.01	6.65	8.58	9.49	9.65	(7)
		HDD50	595	202	124	42	4	0	0	0	0	3	45	175		(8)
		HDD65	2350	597	439	270	94	15	0	0	0	3	79	289	564	(9)
		CDD50	6496	73	114	272	454	727	954	1102	845	535	232	80		(10)
		CDD65	2774	3	8	35	95	277	504	637	643	398	145	25	4	(11)
		CDH74	33486	27	73	285	783	2670	6099	8614	8777	4533	1411	186	28	(12)
		CDH80	17244	2	10	54	178	999	3122	4949	5161	2238	496	32	3	(13)
(14)	Wind	WSAvg	8.6	8.8	9.2	10.0	10.3	9.8	9.0	7.8	7.0	6.7	7.8	8.4	8.3	(14)
(15)	Precipitation	PrecAvg	35.3	2.1	2.1	2.9	3.7	4.4	3.3	2.2	2.3	2.9	4.0	2.9	2.4	(15)
(16)		PrecMax	56.8	5.7	7.0	6.2	7.8	13.5	12.4	5.0	7.2	10.5	10.8	9.5	9.9	(16)
(17)		PrecMin	14.5	0.2	0.0	0.0	0.6	0.5	0.4	0.0	0.6	0.5	0.1	0.4		(17)
(18)		PrecStd	9.6	1.3	1.8	1.7	1.8	2.7	2.8	1.5	1.8	2.2	2.8	2.4	1.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	DB	77.8	81.3	85.3	89.5	96.0	100.9	105.3	106.3	101.8	93.0	83.8	78.6		(19)
(20)		MCWB	58.2	60.3	65.3	69.6	73.5	74.1	73.6	73.5	72.8	71.1	65.8	62.1		(20)
(21)		DB	72.3	75.5	80.8	84.3	92.0	98.1	101.9	103.0	97.1	89.3	79.2	72.4		(21)
(22)		MCWB	57.9	60.5	64.8	67.9	72.9	74.6	74.2	73.9	73.2	70.3	64.4	59.3		(22)
(23)		DB	67.7	71.7	77.0	81.5	88.9	95.4	99.5	100.3	93.8	85.5	75.1	68.3		(23)
(24)		MCWB	56.1	59.6	62.5	67.3	72.2	74.4	75.1	74.4	72.7	68.5	63.3	58.3		(24)
(25)		DB	63.5	67.2	73.1	78.9	85.9	92.8	97.0	98.2	91.1	82.0	71.9	63.7		(25)
(26)		MCWB	54.7	57.4	61.3	65.8	71.3	74.3	74.8	74.3	72.4	67.1	61.5	55.0		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	WB	65.7	67.3	69.7	74.0	78.1	79.8	79.5	78.8	77.8	75.3	70.9	66.5		(27)
(28)		MCDB	70.7	74.8	80.2	82.9	89.2	93.2	94.8	93.6	91.1	84.7	77.8	72.7		(28)
(29)		WB	62.6	65.2	67.5	71.5	76.0	78.1	78.3	77.7	76.4	73.4	68.0	63.8		(29)
(30)		MCDB	67.4	71.1	76.1	79.9	86.1	90.8	93.3	93.2	89.0	83.2	74.5	70.2		(30)
(31)		WB	59.3	62.4	65.8	70.0	74.7	77.1	77.4	76.9	75.3	72.1	65.8	60.0		(31)
(32)		MCDB	65.5	68.3	73.7	77.9	84.4	89.3	92.2	92.2	87.4	81.2	72.5	65.2		(32)
(33)		WB	54.7	59.1	63.8	68.4	73.3	76.0	76.7	76.0	74.4	70.3	63.6	56.1		(33)
(34)		MCDB	62.3	66.1	71.2	75.9	82.8	87.9	91.0	90.9	86.0	78.5	70.1	62.9		(34)
(35)	Mean Daily Temperature Range	MDBR	22.3	21.9	21.9	21.5	19.8	20.0	20.9	21.5	21.8	22.6	21.7	21.1		(35)
(36)		5% DB	29.0	27.5	25.7	23.8	22.4	22.5	23.5	24.2	24.1	25.2	24.4	26.8		(36)
(37)		MCWBR	17.3	16.4	13.7	12.3	9.2	6.6	6.3	6.2	7.4	11.1	13.5	16.3		(37)
(38)		5% WB	23.3	23.2	20.7	18.8	18.9	19.0	20.4	20.7	19.7	19.6	20.4	20.8		(38)
(39)		MCWBR	16.8	16.3	12.8	11.7	9.4	7.2	6.7	6.2	6.9	9.8	14.5	16.0		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.311	0.323	0.355	0.390	0.423	0.435	0.437	0.436	0.398	0.358	0.334	0.316		(40)
(41)		taud	2.499	2.473	2.403	2.331	2.299	2.301	2.333	2.320	2.373	2.470	2.476	2.485		(41)
(42)		Ebn at Noon	288	296	292	285	275	271	269	268	275	279	277	279		(42)
(43)		Edh at Noon	27	31	36	40	42	42	40	40	36	31	28	26		(43)
(44)	All-Sky Solar Radiation	RadAvg	923	1091	1431	1734	1869	2102	2138	1960	1641	1286	978	809		(44)
(45)		RadStd	93	171	108	141	185	200	145	122	143	191	126	102		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	Regional (43 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page